

Accuracy & Precision in Measurement

Nagendra Bahadur Amatya

Associate Professor

nbamatya@ioe.edu.np

9851072341

Accuracy & Precision

- Accuracy:

- How close you are to the actual value
- Depends on the person measuring
- Calculated by the formula:

$$\% \text{ Error} = (YV - AV) \times 100 \div AV$$

Where: YV is YOUR measured Value &
AV is the Accepted Value

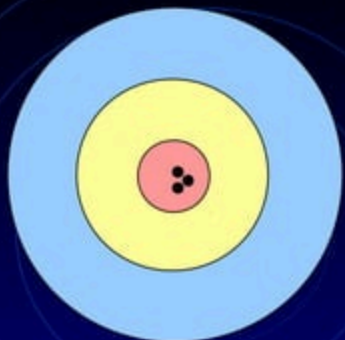
- Precision:

- How finely tuned your measurements are or how close they can be to each other
- Depends on the measuring tool
- Determined by the number of significant digits

Accuracy & Precision

- Accuracy & Precision may be demonstrated by shooting at a target.
- Accuracy is represented by hitting the bulls eye (the accepted value)
- Precision is represented by a tight grouping of shots (they are finely tuned)

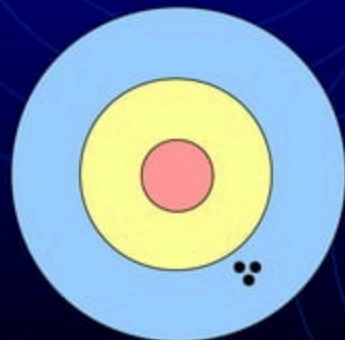
Accuracy & Precision



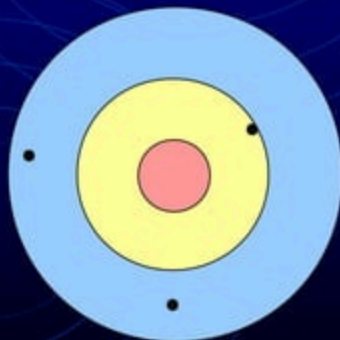
Accuracy without Precision

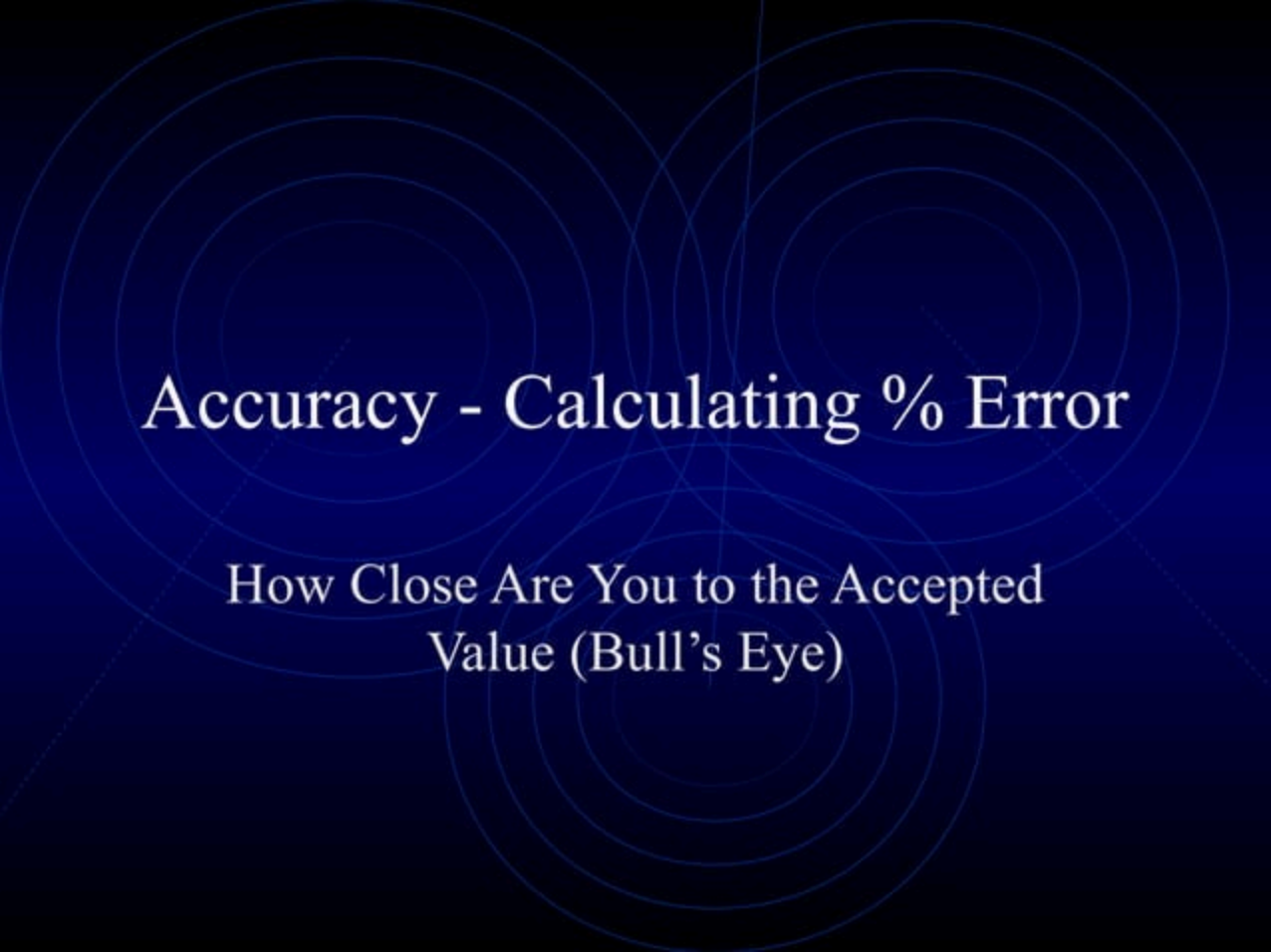


Precision without Accuracy



No Precision & No Accuracy



The background of the slide is dark blue and features several faint, concentric circles. In the center, there is a bullseye target graphic, which is a series of concentric circles with a small dot in the middle, representing the 'Bull's Eye' mentioned in the text.

Accuracy - Calculating % Error

How Close Are You to the Accepted
Value (Bull's Eye)

- **Accuracy is the closeness of a measured value to the true value.**
- For example, the measured density of water has become more accurate with improved experimental design, technique, and equipment.

Density of H₂O at 20° C
(g/cm³)

1

1.0

1.00

0.998

0.9982

0.99820

0.998203

Accuracy - Calculating % Error

- If a student measured the room width at 8.46 m and the **accepted value** was 9.45 m what was their accuracy?
- Using the formula:
$$\% \text{ error} = (YV - AV) \times 100 \div AV$$
 - Where YV is the student's measured value & AV is the **accepted value**

Accuracy - Calculating % Error

- Since $YV = 8.46 \text{ m}$, $AV = 9.45 \text{ m}$
- $\% \text{ Error} = (8.46 \text{ m} - 9.45 \text{ m}) \times 100 \div 9.45 \text{ m}$
- $= -0.99 \text{ m} \times 100 \div 9.45 \text{ m}$
- $= -99 \text{ m} \div 9.45 \text{ m}$
- $= -10.5 \%$
 - Note that the meter unit cancels during the division & the unit is %. The (-) shows that YV was low
- The student was off by almost 11% & must remeasure
- Accepted 11.0% error is 11.0% 50%

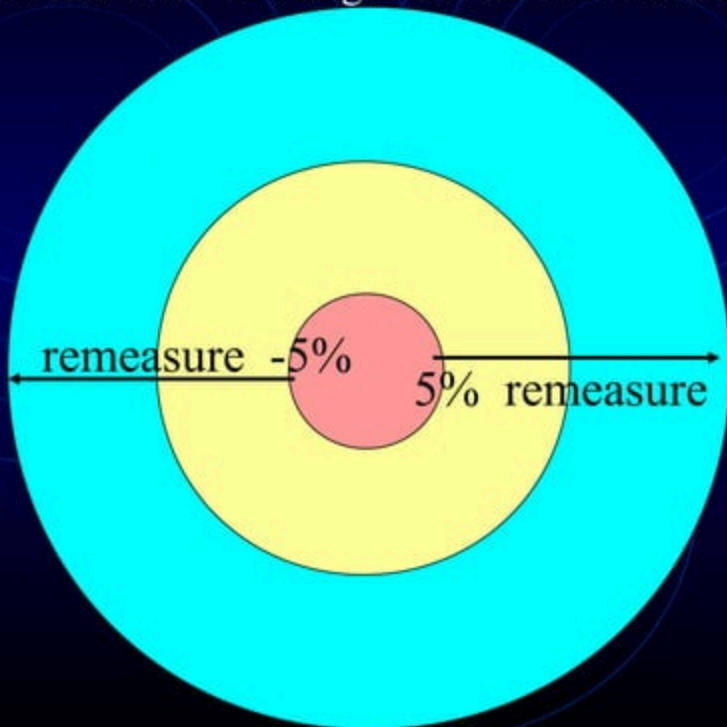
- percent error is used to estimate the accuracy of a surement.

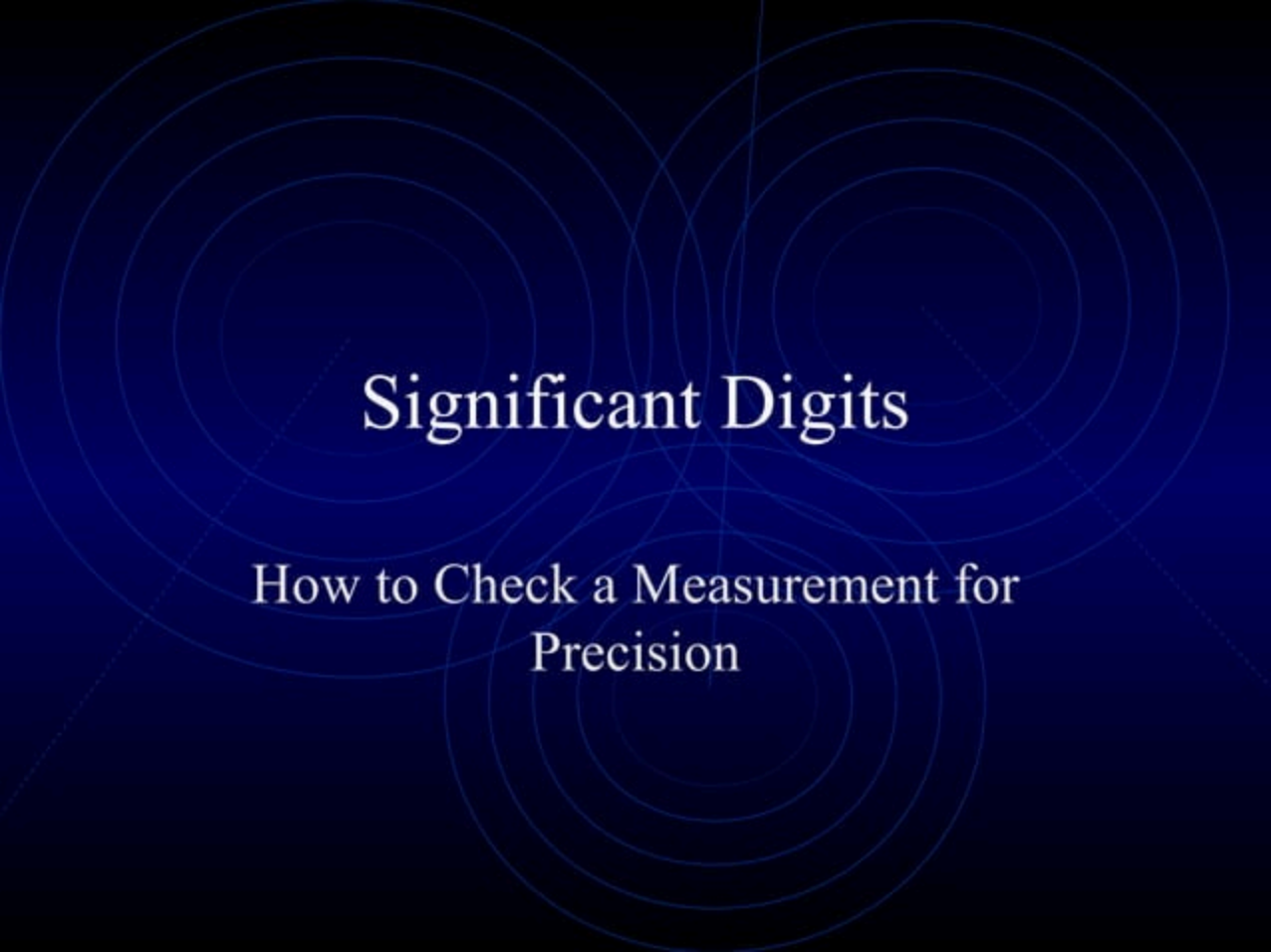
- $$\text{percent error} = \frac{\text{Measured value} - \text{Acceptable value}}{\text{Acceptable value}} \times 100$$

- percent error will always Positive

What is the percent error if the measured density of titanium (Ti) 45 g/cm³ and the accepted density of Ti is 4.50 g/cm³?

- Acceptable error is $\pm 5\%$
- Values from -5% up to 5% are acceptable
- Values less than -5% or greater than 5% must be remeasured



The background is a dark blue gradient. It features several sets of concentric circles in a lighter blue color, centered at different points. Overlaid on these circles is a faint, light blue grid pattern consisting of thin lines forming a rectangular mesh.

Significant Digits

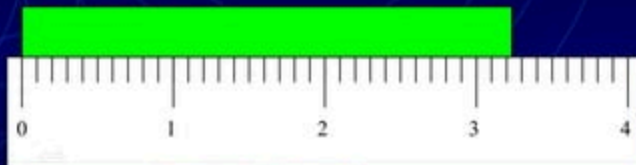
How to Check a Measurement for
Precision

Significant Digits & Precision

- The precision of a measurement is the smallest possible unit that could be measured.
- Significant Digits (sd) are the numbers that result from a measurement.
- When a measurement is converted we need to make sure we know which digits are significant and keep them in our conversion
- All digits that are measured are significant

Significant Digits & Precision

- What is the length of the bar?
- How many digits are there in the measurement?
- All of these digits are significant
- There are 3 sd



Length of Bar = 3.23 cm

Significant Digits & Precision

- If we converted to that measurement of 3.23 cm to μm what would we get?
- Right! 32 300 μm
- How many digits in our converted number?
- Are they all significant digits (measured)?
- Which ones were measured and which ones were added because we converted?
- If we know the significant digits we can know the precision of our original measurement

Significant Digits & Precision

- What if we didn't know the original measurement – such as 0.005670 hm. How would we know the precision of our measurement.
- The rules showing how to determine the number of significant digits is shown in your lab manual on p. 19. Though you can handle them, they are somewhat complex.

Types of Errors

Difference between measured result and true value.

- Illegitimate errors

- Blunders resulting from mistakes in procedure. You must be careful.
- Computational or calculational errors after the experiment.

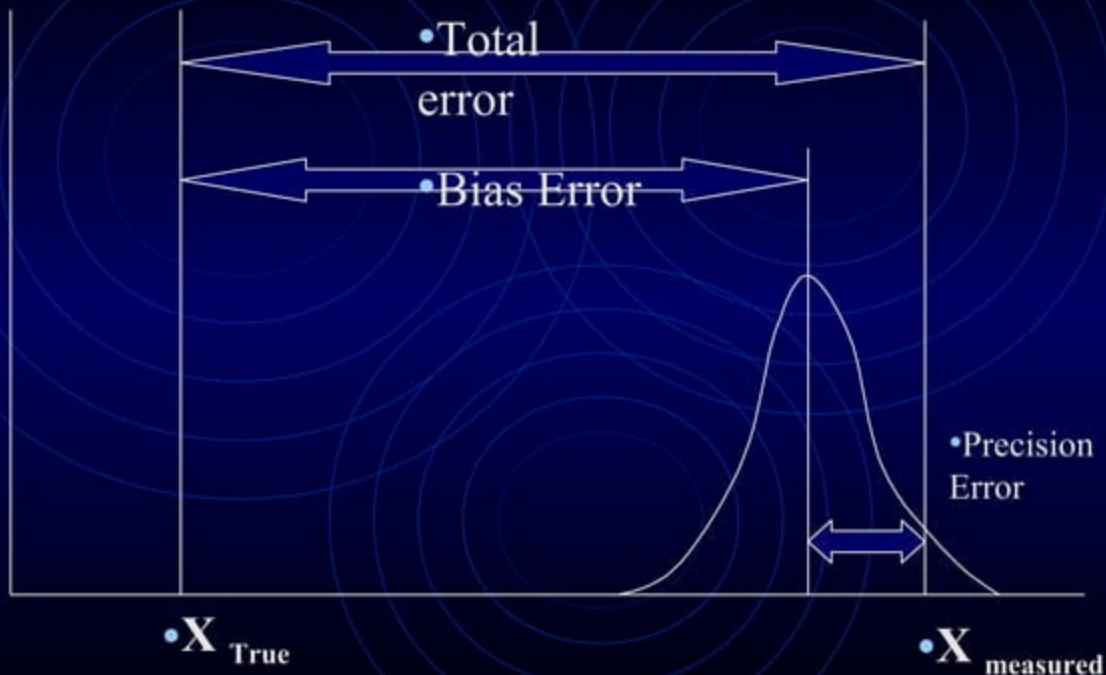
- Bias or Systematic errors

- An offset error; one that remains with repeated measurements (i.e. a change of indicated pressure with the difference in temperature from calibration to use).
 - **Systematic errors can be reduced through calibration**
 - **Faulty equipment--such as an instrument which always reads 3% high**
 - **Consistent or recurring human errors-- observer bias**
- This type of error cannot be evaluated directly from the data but can be determined by comparison to theory or other experiments.

Types of Errors (cont.)

- Random, Stochastic or Precision errors:
 - An error that causes readings to take random-like values about the mean value.
 - Effects of uncontrolled variables
 - Variations of procedure
 - The concepts of probability and statistics are used to study random errors. When we think of random errors we also think of repeatability or precision.

Bias, Precision, and Total Error



Accuracy and Precision

- Accuracy is the closeness of a measurement (or set of observations) to the true value. The higher the accuracy the lower the error.
- Precision is the closeness of multiple observations to one another, or the repeatability of a measurement.

Uncertainty Analysis

- The estimate of the error is called the uncertainty.
 - It includes both bias and precision errors.
 - We need to identify all the potential significant errors for the instrument(s).
 - All measurements should be given in three parts
 - Mean value
 - Uncertainty
 - Confidence interval on which that uncertainty is based (typically 95% C.I.)
- Uncertainty can be expressed in either absolute terms (i.e., 5 Volts ± 0.5 Volts) or in percentage terms (i.e., 5 Volts $\pm 10\%$) (relative uncertainty = $\Delta V / V \times 100$)
- We will use a 95 % confidence interval throughout this course (20:1 odds).

How to Estimate Bias Error

- Manufacturers' Specifications
 - If you can't do better, you may take it from the manufacturer's specs.
 - Accuracy - %FS, %reading, offset, or some combination (e.g., 0.1% reading + 0.15 counts)
 - Unless you can identify otherwise, assume that these are at a 95% confidence interval
- Independent Calibration
 - May be deduced from the calibration process

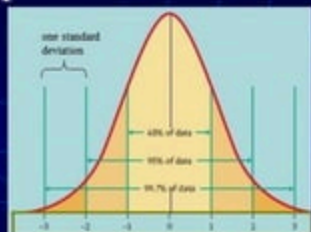
Use Statistics to Estimate Random Uncertainty

- Mean: the sum of measurement values divided by the number of measurements.
- Deviation: the difference between a single result and the mean of many results.
- Standard Deviation: the smaller the standard deviation the more precise the data
 - Large sample size →
 - Small sample size ($n < 30$)
 - Slightly larger value →

The Population

- Population: The collection of all items (measurements) of the group. Represented by a large number of measurements.

- Gaussian distribution*



- Sample: A portion of (or limited number of items in) a population.
- **Data do not always abide by the Gaussian distribution. If not, you must use another method!!*

Student t-distribution (small sample sizes)

- The t-distribution was formulated by W.S. Gosset, a scientist in the Guinness brewery in Ireland, who published his formulation in 1908 under the pen name (pseudonym) "*Student*."
- The t-distribution looks very much like the Gaussian distribution, bell shaped, symmetric and centered about the mean. The primary difference is that it has stronger tails, indicating a lower probability of being within an interval. The variability depends on the sample size, n .

Student t-distribution

- With a confidence interval of c%

$$\bar{X} - t_{\alpha/2, v} \frac{\sigma_s}{\sqrt{n}} < X < \bar{X} + t_{\alpha/2, v} \frac{\sigma_s}{\sqrt{n}}$$

- Where $\alpha=1-c$ and $v=n-1$ (Degrees of Freedom)
- Don't apply blindly - you may have better information about the population than you think.

Example: t-distribution

- Sample data
 - $n = 21$
 - Degrees of Freedom = $\nu = 20$
- Desire 95% Confidence Interval
 - $\forall \alpha = 1 - c = 0.05$
 - $\forall \alpha/2 = 0.025$
- Student t-dist chart
 - $t = 2.086$

Estimate of Precision Error is Then:

$$\bar{x} \pm t_{\alpha/2, n} \cdot \frac{\sigma}{\sqrt{n}}$$

$$5.60 \pm 2.086 \cdot \frac{0.51}{\sqrt{21}}$$

$$5.60 \pm 0.23$$

- Precision error is
 - ± 0.23 Volts

How to combine bias and precision error?

- Rules for combining independent uncertainties for measurements:
 - Both uncertainties **MUST** be at the same Confidence Interval (95%)

$$U_x = \sqrt{B_x^2 + P_x^2}$$

- Precision error obtained using Student's-t method
- Bias error determined from calibration, manufacturers' specifications, smallest division.

Propagation of Error

- Used to determine uncertainty of a quantity that requires measurement of several independent variables.
 - Volume of a cylinder = $f(D,L)$
 - Volume of a block = $f(L,W,H)$
 - Density of an ideal gas = $f(P,T)$
- Again, all variables must have the same confidence interval.

RSS Method (Root Sum Squares)

- For a function $y = f(x_1, x_2, \dots, x_N)$, the RSS uncertainty is:

$$\Delta y_{RSS} = \sqrt{\left[\frac{\partial f}{\partial x_1} \Delta x_1 \right]^2 + \left[\frac{\partial f}{\partial x_2} \Delta x_2 \right]^2 + \dots + \left[\frac{\partial f}{\partial x_N} \Delta x_N \right]^2}$$

- First determine uncertainty of each variable in the form $(x_N \pm \Delta x_N)$
 - Use previously established methods, including bias and precision error.

Example Problem: Propagation of Error

- Ideal gas law:
- Temperature
 - $T \pm \Delta T$
- Pressure
 - $P \pm \Delta P$
- $R = \text{Constant}$

$$\rho = \frac{P}{RT}$$

- **How do we estimate the error in the density?**

- Apply RSS formula to density relationship:

$$\Delta \rho_{\text{RSS}} = \sqrt{\left[\frac{\partial \rho}{\partial P} \Delta P \right]^2 + \left[\frac{\partial \rho}{\partial T} \Delta T \right]^2} = \sqrt{\left[\frac{1}{R T} \Delta P \right]^2 + \left[\frac{-P}{R T^2} \Delta T \right]^2}$$

- Apply a little algebra:

$$\rho = \frac{P}{RT}$$

$$\frac{\Delta \rho}{\rho} = \sqrt{\left[\frac{\Delta P}{P} \right]^2 + \left[\frac{\Delta T}{T} \right]^2}$$